

Table 1: Performance and computational efficiency of Qwen3-8B under the sequential configuration. Acc. ($\uparrow$) denotes task accuracy (%), or pass@1 (%) for code generation; Token ($\downarrow$) denotes tokens per problem; and Time ($\downarrow$) denotes inference time (seconds per problem). Single, Text, and Linear Align times are converted to the Kernel machine’s throughput. AVG denotes the unweighted mean across all six benchmarks. The highest accuracy among training-free latent alignment methods, the lowest token count, and the shortest inference time in each column (including AVG) are shown in bold.

Method	Metric	AIME 2024	AIME 2025	Human Eval+	MBPP+	GPQA- Diamond	MedQA	AVG
Single	Acc.	50.00	46.70	74.40	64.80	39.90	53.00	54.80
	Token	12891	14692	2507	2053	6435	2098	6779.33
	Time	340.2	377.8	31.5	28.9	104.6	18.3	150.22
Text	Acc.	53.30	53.30	80.50	69.50	43.40	75.00	62.50
	Token	38596	45088	4593	3695	17986	4260	19036.33
	Time	1018.6	1159.4	57.7	52.1	292.2	37.1	436.18
Linear Align (?)	Acc.	56.70	53.30	80.50	74.60	45.50	75.30	64.32
	Token	8953	8699	1866	1164	4571	1555	4468.00
	Time	236.3	223.7	23.4	16.4	74.3	13.5	97.93
Exact-Soft Align (?)	Acc.	62.22 $\pm$ 6.85	52.22 $\pm$ 8.31	83.13 $\pm$ 2.24	72.49 $\pm$ 0.75	48.48 $\pm$ 1.24	68.67 $\pm$ 1.44	64.54
	Token	22496 $\pm$ 1178	26193 $\pm$ 1716	14598 $\pm$ 308	14297 $\pm$ 370	13058 $\pm$ 315	15480 $\pm$ 383	17686.90
	Time	575.3 $\pm$ 33.7	634.6 $\pm$ 18.5	107.1 $\pm$ 7.4	101.6 $\pm$ 6.9	257.4 $\pm$ 10.7	107.6 $\pm$ 2.7	297.25
Kernel Align	Acc.	68.89$\pm$1.57	54.44 $\pm$ 4.16	83.94$\pm$0.57	75.40 $\pm$ 0.22	49.16 $\pm$ 1.04	73.22 $\pm$ 0.87	67.51
	Token	8548$\pm$344	10230 $\pm$ 286	1639$\pm$42	1463 $\pm$ 19	3574$\pm$87	1163 $\pm$ 20	4436.13
	Time	229.3 $\pm$ 3.3	271.2 $\pm$ 5.8	18.9$\pm$0.2	18.1 $\pm$ 0.2	57.6$\pm$0.6	13.3$\pm$0.5	101.39
Kernel-ES Align	Acc.	65.56 $\pm$ 1.57	57.78$\pm$4.16	83.94$\pm$1.60	75.49$\pm$0.50	50.17$\pm$1.04	73.22 $\pm$ 1.29	67.69
	Token	8552 $\pm$ 586	9419 $\pm$ 427	1692 $\pm$ 34	1532 $\pm$ 11	3865 $\pm$ 77	1148$\pm$8	4367.97
	Time	216.0$\pm$13.2	244.2 $\pm$ 18.6	21.5 $\pm$ 0.4	19.6 $\pm$ 0.4	60.0 $\pm$ 0.6	13.4 $\pm$ 0.1	95.79

Table 2: Performance and computational efficiency of Qwen3-8B under the hierarchical configuration. Acc. ($\uparrow$) denotes task accuracy (%), or pass@1 (%) for code generation; Token ($\downarrow$) denotes tokens per problem; and Time ($\downarrow$) denotes inference time (seconds per problem). Single, Text, and Linear Align times are converted to the Kernel machine’s throughput. AVG denotes the unweighted mean across all six benchmarks. The highest accuracy among training-free latent alignment methods, the lowest token count, and the shortest inference time in each column (including AVG) are shown in bold. A dash indicates an unavailable result; AVG is reported only when all six benchmark results are available.

Method	Metric	AIME 2024	AIME 2025	Human Eval+	MBPP+	GPQA- Diamond	MedQA	AVG
Single	Acc.	50.00	46.70	74.40	64.80	39.90	53.00	54.80
	Token	12891	14692	2507	2053	6435	2098	6779.33
	Time	335.6	379.0	30.7	33.8	102.7	25.1	151.15
Text	Acc.	53.30	50.00	76.80	71.90	43.00	76.30	61.88
	Token	42629	53929	8768	7703	22450	6893	23728.67
	Time	1109.7	1391.2	107.5	126.8	358.4	82.5	529.35
Linear Align (?)	Acc.	53.30	50.00	78.00	72.20	46.90	77.00	62.90
	Token	7526	13230	1274	1264	3395	1007	4616.00
	Time	195.9	341.3	15.6	20.8	54.2	12.1	106.65
Exact-Soft Align (?)	Acc.	–	55.56 $\pm$ 6.29	–	73.72$\pm$0.25	–	–	–
	Token	–	29403 $\pm$ 987	–	12809 $\pm$ 472	–	–	–
	Time	–	699.0 $\pm$ 32.0	–	92.1 $\pm$ 5.7	–	–	–
Kernel Align	Acc.	65.56 $\pm$ 1.57	58.89$\pm$3.14	84.35$\pm$0.76	73.02 $\pm$ 0.86	52.36$\pm$2.34	73.33	67.92
	Token	9102 $\pm$ 389	11207 $\pm$ 253	1697 $\pm$ 27	1495 $\pm$ 13	3728 $\pm$ 106	1351	4763.12
	Time	240.8 $\pm$ 5.2	292.3 $\pm$ 10.8	20.5 $\pm$ 0.4	20.0$\pm$0.0	59.4 $\pm$ 2.0	15.7	108.14
Kernel-ES Align	Acc.	68.89$\pm$3.14	56.67 $\pm$ 2.72	83.74 $\pm$ 1.04	73.10 $\pm$ 0.66	50.67 $\pm$ 0.86	72.67	67.62
	Token	9805 $\pm$ 55	11143$\pm$323	1609 $\pm$ 119	1543 $\pm$ 16	3913 $\pm$ 67	1336	4891.43
	Time	264.4 $\pm$ 2.8	288.3$\pm$5.9	20.9 $\pm$ 1.0	20.6 $\pm$ 0.2	60.7 $\pm$ 1.4	14.8	111.61

Table 3: Performance and computational efficiency of Qwen3-14B under the sequential configuration. Acc. ($\uparrow$) denotes task accuracy (%), or pass@1 (%) for code generation; Token ($\downarrow$) denotes tokens per problem; and Time ($\downarrow$) denotes inference time (seconds per problem). Single, Text, and Linear Align times are converted to the Kernel machine’s throughput. AVG denotes the unweighted mean across all six benchmarks. The highest accuracy among training-free latent alignment methods, the lowest token count, and the shortest inference time in each column (including AVG) are shown in bold. A dash indicates an unavailable result; AVG is reported only when all six benchmark results are available.

Method	Metric	AIME 2024	AIME 2025	Human Eval+	MBPP+	GPQA- Diamond	MedQA	AVG
Single	Acc.	63.30	56.70	76.80	68.50	48.50	64.70	63.08
	Token	11263	11298	2366	1858	5547	1746	5679.67
	Time	348.6	369.2	31.8	28.5	109.5	21.1	151.45
Text	Acc.	63.30	60.00	81.10	72.80	51.50	80.30	68.17
	Token	32092	44618	5934	4971	12676	3444	17289.17
	Time	993.2	1458.1	79.7	76.2	250.2	41.6	483.17
Linear Align (?)	Acc.	66.70	63.30	86.50	75.70	52.00	80.70	70.82
	Token	10593	11402	2042	1621	5454	1841	5492.17
	Time	327.8	372.6	27.4	24.9	107.7	22.2	147.10
Exact-Soft Align (?)	Acc.	68.89 $\pm$ 5.67	54.44 $\pm$ 3.14	91.26$\pm$1.44	–	52.53 $\pm$ 1.24	79.00	–
	Token	19493 $\pm$ 709	20402 $\pm$ 151	19784 $\pm$ 655	–	18443 $\pm$ 848	22875	–
	Time	515.1 $\pm$ 9.8	519.1 $\pm$ 13.7	158.2 $\pm$ 7.1	–	229.7 $\pm$ 21.1	151.0	–
Kernel Align	Acc.	75.56$\pm$3.14	60.00 $\pm$ 2.72	88.21 $\pm$ 0.57	78.13 $\pm$ 0.12	56.23$\pm$0.48	81.67	73.30
	Token	8045$\pm$310	10076 $\pm$ 314	1598$\pm$42	1339$\pm$18	2999 $\pm$ 55	922	4163.18
	Time	243.3$\pm$5.9	328.7$\pm$9.6	24.1$\pm$0.8	21.4$\pm$0.2	59.2$\pm$1.8	13.2	114.98
Kernel-ES Align	Acc.	74.44 $\pm$ 3.14	64.44$\pm$1.57	86.79 $\pm$ 1.25	79.72$\pm$0.76	55.72 $\pm$ 0.24	79.33	73.41
	Token	8360 $\pm$ 96	10072$\pm$549	1605 $\pm$ 26	1352 $\pm$ 8	2972$\pm$86	912	4212.07
	Time	276.4 $\pm$ 2.4	341.1 $\pm$ 24.4	28.4 $\pm$ 0.3	25.7 $\pm$ 0.1	66.2 $\pm$ 2.3	14.3	125.33

Table 4: Performance and computational efficiency of Qwen3-14B under the hierarchical configuration. Acc. ($\uparrow$) denotes task accuracy (%), or pass@1 (%) for code generation; Token ($\downarrow$) denotes tokens per problem; and Time ($\downarrow$) denotes inference time (seconds per problem). Single, Text, and Linear Align times are converted to the Kernel machine’s throughput. AVG denotes the unweighted mean across all six benchmarks. The highest accuracy among training-free latent alignment methods, the lowest token count, and the shortest inference time in each column (including AVG) are shown in bold. A dash indicates an unavailable result; AVG is reported only when all six benchmark results are available.

Method	Metric	AIME 2024	AIME 2025	Human Eval+	MBPP+	GPQA- Diamond	MedQA	AVG
Single	Acc.	63.30	56.70	76.80	68.50	48.50	64.70	63.08
	Token	11263	11298	2366	1858	5547	1746	5679.67
	Time	357.6	364.4	26.4	16.3	123.2	31.6	153.25
Text	Acc.	70.00	66.70	84.10	73.00	52.00	78.00	70.63
	Token	29025	50003	8114	7458	20931	5473	20167.33
	Time	921.7	1612.8	90.6	65.4	464.9	99.0	542.40
Linear Align (?)	Acc.	73.30	66.70	86.60	73.80	53.00	78.30	71.95
	Token	10230	9527	1512	1187	3606	899	4493.50
	Time	324.8	307.3	16.9	10.4	80.1	16.3	125.97
Exact-Soft Align (?)	Acc.	71.11 $\pm$ 5.67	62.22 $\pm$ 6.29	90.85$\pm$1.72	–	54.38 $\pm$ 3.60	48.33	–
	Token	23892 $\pm$ 660	24307 $\pm$ 148	21263 $\pm$ 1076	–	21220 $\pm$ 530	23700	–
	Time	640.9 $\pm$ 33.3	689.2 $\pm$ 11.8	185.4 $\pm$ 12.7	–	296.5 $\pm$ 4.6	173.7	–
Kernel Align	Acc.	72.22 $\pm$ 1.57	64.44 $\pm$ 1.57	87.60 $\pm$ 1.25	78.66$\pm$0.90	56.06 $\pm$ 1.80	79.00	73.00
	Token	8844$\pm$104	10817 $\pm$ 88	1439 $\pm$ 19	1133$\pm$10	2749 $\pm$ 69	860	4307.07
	Time	282.7$\pm$2.9	346.2 $\pm$ 8.1	23.7 $\pm$ 0.4	21.6 $\pm$ 0.2	61.2$\pm$0.9	12.4	124.63
Kernel-ES Align	Acc.	72.22 $\pm$ 1.57	63.33 $\pm$ 4.71	87.40 $\pm$ 0.29	78.66 $\pm$ 0.54	56.40$\pm$0.86	77.67	72.61
	Token	9336 $\pm$ 214	10874 $\pm$ 214	1421$\pm$5	1181 $\pm$ 40	2724$\pm$15	875	4401.93
	Time	302.7 $\pm$ 20.1	348.9 $\pm$ 24.5	25.1 $\pm$ 1.1	23.4 $\pm$ 1.0	62.0 $\pm$ 0.7	13.8	129.33

Table 5: Performance and computational efficiency of cross-model communication between Qwen3-8B and Qwen3-14B. Arrows indicate the sender-to-receiver direction. Acc. ($\uparrow$) denotes task accuracy (%), or pass@1 (%) for code generation; Dec. ($\downarrow$) denotes explicitly decoded vocabulary tokens per problem; and Time ($\downarrow$) denotes end-to-end inference time (seconds per problem). Receiver-only answers the original question without sender representations, using the same receiver generation settings. AVG denotes the unweighted mean across all six benchmarks; it is omitted when any benchmark is unavailable. Values are reported as mean $\pm$ population standard deviation across repeats. A dash indicates an unavailable result.

Method	Metric	AIME 2024	AIME 2025	Human Eval+	MBPP+	GPQA- Diamond	MedQA	AVG
<i>Qwen3-14B $\rightarrow$ Qwen3-8B</i>								
Text	Acc.	68.89 $\pm$ 1.57	58.89 $\pm$ 1.57	92.07 $\pm$ 0.50	82.01 $\pm$ 0.57	59.60 $\pm$ 1.43	80.67 $\pm$ 0.27	73.69 $\pm$ 0.52
	Dec.	9793 $\pm$ 711	10716 $\pm$ 101	2595 $\pm$ 16	2338 $\pm$ 24	4227 $\pm$ 53	1549 $\pm$ 34	5203 $\pm$ 100
	Time	333.0 $\pm$ 45.3	346.1 $\pm$ 17.9	31.1 $\pm$ 0.9	31.8 $\pm$ 0.4	70.7 $\pm$ 0.4	14.9 $\pm$ 0.5	137.9 $\pm$ 7.5
Linear Align	Acc.	2.22 $\pm$ 1.57	63.33 $\pm$ 5.44	8.33 $\pm$ 1.52	21.34 $\pm$ 0.76	18.01 $\pm$ 1.19	77.00 $\pm$ 1.19	31.71 $\pm$ 1.10
	Dec.	2014 $\pm$ 267	12065 $\pm$ 466	1518 $\pm$ 69	1761 $\pm$ 29	2129 $\pm$ 86	1403 $\pm$ 15	3482 $\pm$ 61
	Time	67.7 $\pm$ 14.1	289.9 $\pm$ 6.6	18.2 $\pm$ 1.1	18.1 $\pm$ 0.4	36.5 $\pm$ 1.1	13.0 $\pm$ 0.3	73.9 $\pm$ 1.2
Exact-Soft Align	Acc.	70.00 $\pm$ 2.72	54.44 $\pm$ 3.14	81.91 $\pm$ 2.07	74.96 $\pm$ 0.33	50.84 $\pm$ 2.08	77.67 $\pm$ 0.98	68.30 $\pm$ 0.85
	Dec.	10505 $\pm$ 490	12369 $\pm$ 236	1956 $\pm$ 47	1654 $\pm$ 22	4481 $\pm$ 69	1390 $\pm$ 36	5393 $\pm$ 85
	Time	263.3 $\pm$ 5.7	300.1 $\pm$ 2.3	17.8 $\pm$ 0.5	16.8 $\pm$ 0.4	57.0 $\pm$ 0.7	13.1 $\pm$ 0.6	111.4 $\pm$ 1.3
Kernel Align	Acc.	42.22 $\pm$ 8.31	61.11 $\pm$ 4.16	75.20 $\pm$ 0.57	72.49 $\pm$ 1.42	46.13 $\pm$ 2.12	76.78 $\pm$ 1.29	62.32 $\pm$ 2.23
	Dec.	13991 $\pm$ 412	11802 $\pm$ 517	2355 $\pm$ 33	1727 $\pm$ 41	4185 $\pm$ 133	1418 $\pm$ 12	5913 $\pm$ 118
	Time	321.6 $\pm$ 1.3	283.4 $\pm$ 17.9	20.8 $\pm$ 0.1	17.6 $\pm$ 0.3	53.7 $\pm$ 1.0	13.7 $\pm$ 0.5	118.5 $\pm$ 2.9
<i>Qwen3-8B $\rightarrow$ Qwen3-14B</i>								
Text	Acc.	72.22 $\pm$ 6.85	60.00 $\pm$ 5.44	—	—	—	—	—
	Dec.	10074 $\pm$ 449	12116 $\pm$ 369	—	—	—	—	—
	Time	307.3 $\pm$ 13.3	382.6 $\pm$ 25.4	—	—	—	—	—
Linear Align	Acc.	62.22 $\pm$ 5.67	66.67 $\pm$ 5.44	—	—	—	—	—
	Dec.	10515 $\pm$ 1026	11622 $\pm$ 438	—	—	—	—	—
	Time	331.1 $\pm$ 48.7	364.2 $\pm$ 6.4	—	—	—	—	—
Exact-Soft Align	Acc.	76.67 $\pm$ 5.44	68.33 $\pm$ 5.00	—	—	—	—	—
	Dec.	9678 $\pm$ 328	11840 $\pm$ 236	—	—	—	—	—
	Time	289.5 $\pm$ 21.6	372.5 $\pm$ 8.4	—	—	—	—	—
Kernel Align	Acc.	67.78 $\pm$ 4.16	71.67 $\pm$ 1.67	—	—	—	—	—
	Dec.	11180 $\pm$ 324	11756 $\pm$ 146	—	—	—	—	—
	Time	346.4 $\pm$ 9.7	351.9 $\pm$ 8.5	—	—	—	—	—